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CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.8 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water		=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water		=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice		=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water		=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	s	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$

SECTION A

Attempt ALL questions.

You MUST write your answers in this answer booklet.

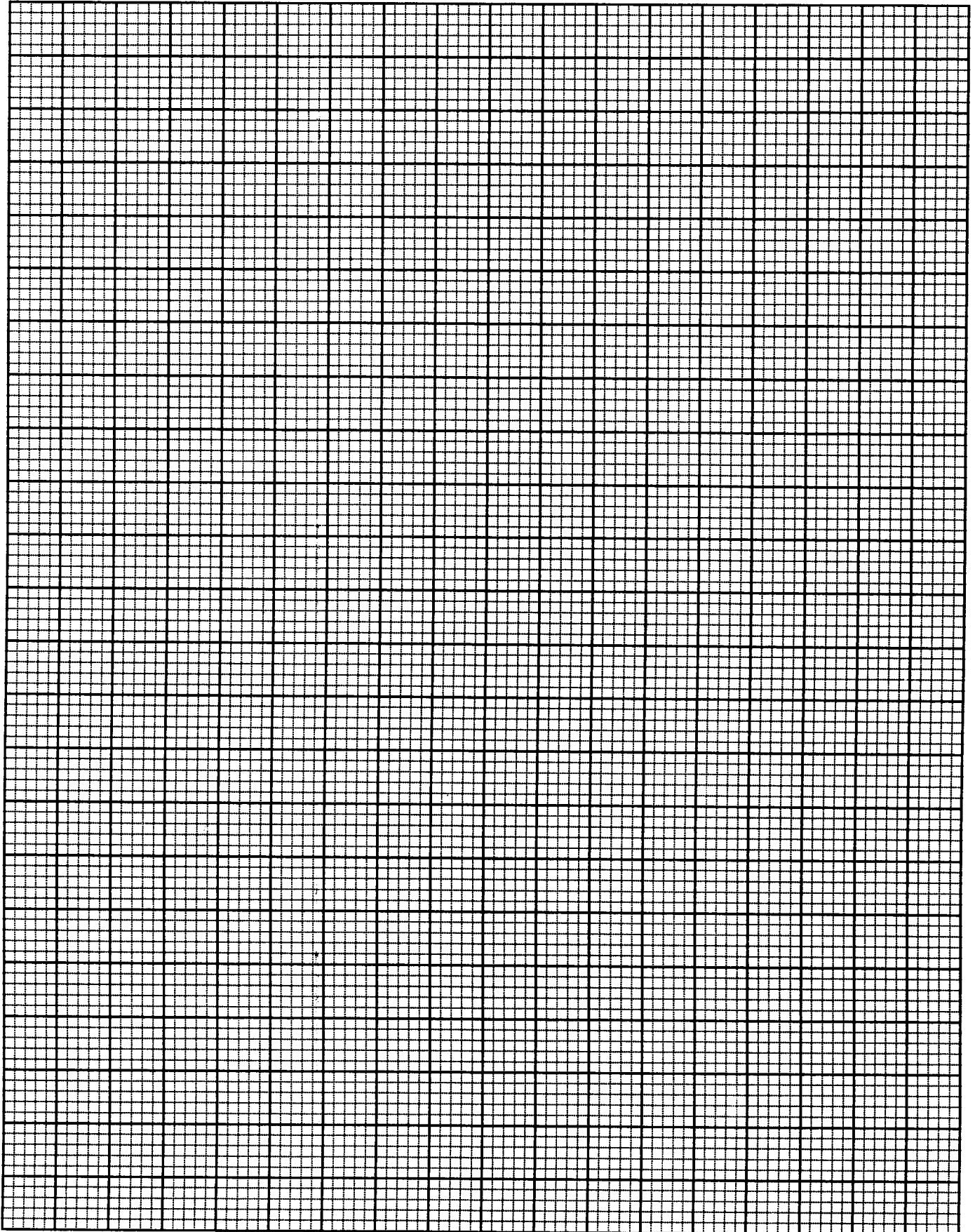
1. (a) Using electronic timing the following data in Table 1 was obtained for a steel ball bearing falling from rest. Use the data to plot a suitable straight line graph (on page 5) to determine the value of, g , the acceleration of free fall. **[6 marks]**

TABLE 1

Distance, y/m	Time, t/s	t^2/s^2
$\pm 2 \text{ mm}$	$\pm 2 \text{ ms}$	—
0.400	0.281	
0.600	0.342	
0.800	0.414	
1.000	0.456	
1.200	0.500	
1.400	0.534	

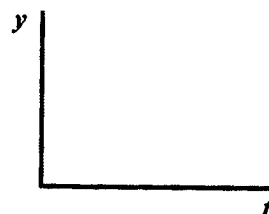
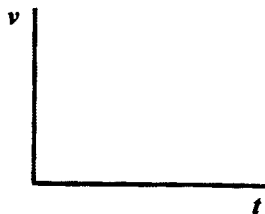
Working for determination of g .

[3 marks]



GO ON TO THE NEXT PAGE

- (b) On the axes below **sketch** graphs (using down as positive) to show how the acceleration, a , velocity, v , and displacement, y , of the falling steel bearing vary with time.



[3 marks]

- (c) Using $g = 9.8 \text{ m s}^{-2}$ rather than the value from (a), find the velocity of the steel bearing after it has fallen a distance of 0.90 m.

[3 marks]

Total 15 marks

2. (a) (i) What is meant by the term 'diffraction'?

[1 mark]

- (ii) Plane waves incident on small apertures are represented in Figure 1 below. Show on the diagram, the resulting waves after passage through the slits.

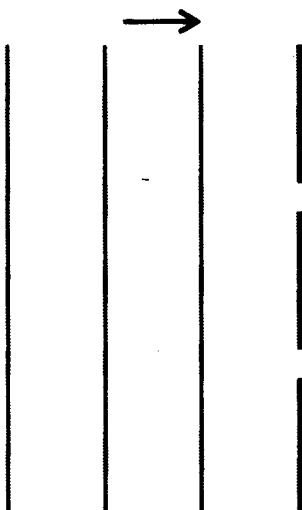


Figure 1

[2 marks]

- (iii) Mark on Figure 1, TWO places where constructive interference occurs and TWO places where destructive interference occurs. [Use a circle O for constructive interference and an X for destructive interference.] [2 marks]
- (iv) In the space below draw wave diagrams to show how TWO waves could superpose to produce constructive and destructive interference. [2 marks]

- (b) Two vertical dipole antennas, A and B , emit coherent signals from a radio station broadcasting at a frequency of 900 kHz as shown in Figure 2. The antennas are spaced 600 m apart.

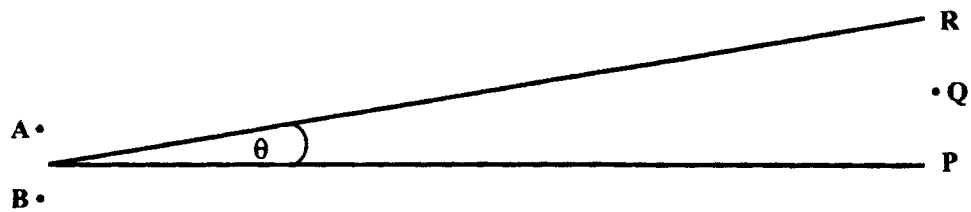


Figure 2

- (i) A maximum intensity of the signal is received at P some distance away on the line perpendicular to AB as shown and the next maximum is detected at R on a line at an angle θ . Calculate the value of θ .

[5 marks]

- (ii) Explain why at Q , a point midway between P and R , the signal received is very weak.

[1 mark]

- (iii) At what angle to the mid-line would the next minimum of the interference pattern occur?

[2 marks]

Total 15 marks

3. (a) (i) Explain what is meant by EACH of the following terms.

'Stress':

[1 mark]

'Strain':

[1 mark]

- (ii) Figure 3 shows a graph of applied force F , against extension, Δl , for a long steel wire. Mark on the graph the region where the wire obeys Hooke's law.

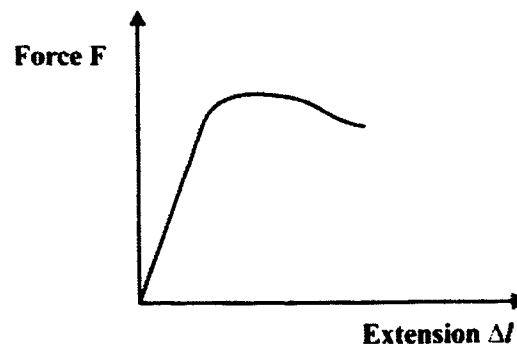


Figure 3

[1 mark]

- (iii) In the space below, sketch graphs to show how the graph in Figure 3 will change if the material is

(a) Glass

(b) Rubber

[2 marks]

- (b) Figure 4 shows a 0.5 kg mass on a steel wire 1 m long and 2×10^{-3} m in diameter (Young modulus = 2×10^{11} Pa) being whirled in a vertical circle. The mass is at its lowest position and its velocity at this point is 38 m s^{-1} . The breaking stress of the wire is $7.2 \times 10^8 \text{ N m}^{-2}$.

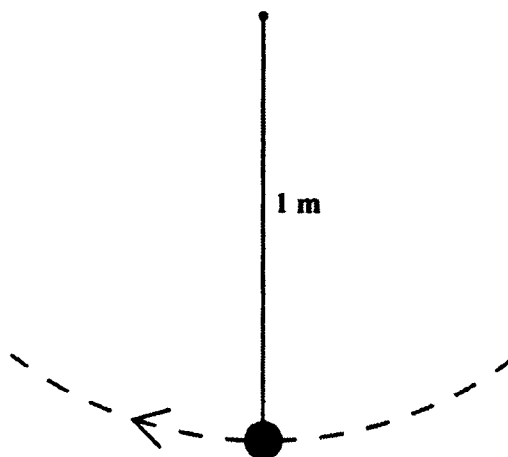


Figure 4

- (i) On Figure 4 show the forces acting on the mass.

[2 marks]

(ii) At the LOWEST point, calculate

a) the tension in the wire

[2 marks]

b) the stress experienced by the wire

[2 marks]

c) the extension of the wire.

[3 marks]

(c) Explain whether or not the wire breaks at this point.

[1 mark]

Total 15 marks

SECTION B

Attempt ALL questions.

You MUST write your answers in the answer booklet provided.

4. (a) Whilst doing an experiment on the stretching of a thin wire (diameter less than 0.5 mm) a student is required to measure its diameter. Describe how the diameter of the wire could be measured accurately. [3 marks]
- (b) Discuss, using examples, the difference between precision and accuracy in physical measurement. [4 marks]
- (c) When a liquid is boiled using an electrical heater the mass of liquid evaporated, m , is related to the power of the heater, P , and the elapsed time, t , by the equation $Pt = mL$. The specific latent heat, L , may be determined if the other quantities are measured. Table 2 shows the results of such an experiment.

TABLE 2

Power of heater	$54 \pm 2 \text{ W}$
Mass of liquid evaporated	$(9.9 \pm 0.1) \times 10^{-3} \text{ kg}$
Time elapsed	$300 \pm 2 \text{ s}$

Use this data to find the specific latent heat of the liquid, together with its uncertainty, and write the answer in the form $L = (x \pm y) \times 10^6 \text{ J kg}^{-1}$, using an appropriate number of significant figures. [8 marks]

Total 15 marks

5. (a) A piece of perspex is placed in a ripple tank to make the water shallower, see Figure 5 (a).

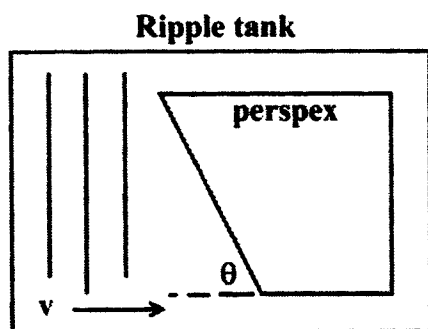


Figure 5 (a)

As the waves cross the front of the perspex, which is at an angle θ to the direction of propagation, they slow down. With the aid of a diagram, explain what happens to the waves. What is this phenomenon called? [3 marks]

- (b) In an experiment with water waves in a ripple tank, a variable frequency a.c. source was used to vibrate a dipper so that it produced ripples on the water surface. The wavelength, λ , and wave speed, v , of the ripples were measured for various dipper frequencies.

The graph in Figure 5 (b) shows the variation of wave speed, v , with wavelength, λ , for ripples up to a wavelength of 8 mm.

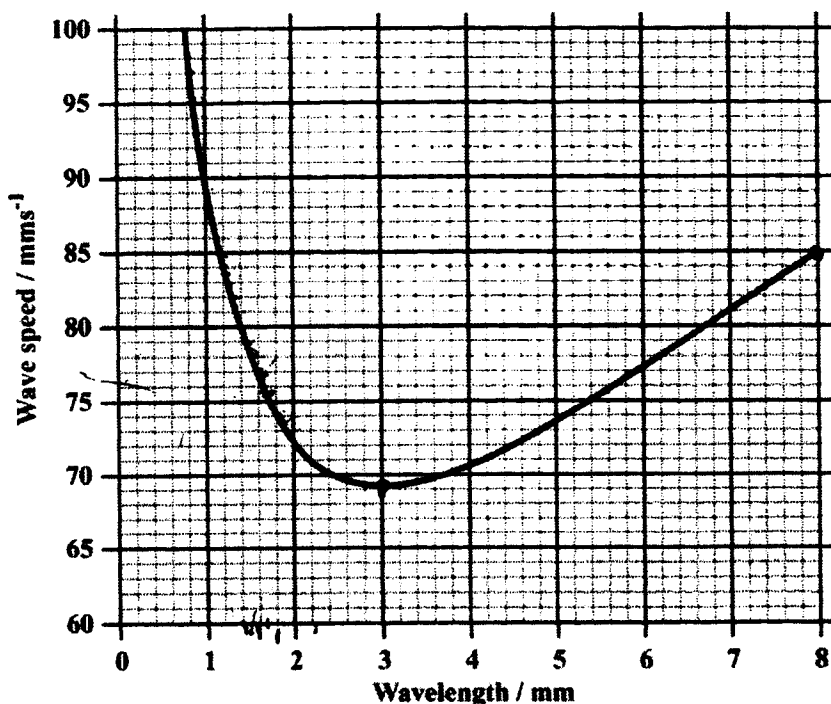


Figure 5 (b)

- (i) By how much does the frequency decrease when the wavelength is changed from 3 mm to 8 mm? **[2 marks]**

- (ii) With water of the depth used in the experiment, it can be shown that the speed of the ripples of **wavelength less than 2.0 mm** is given by

$$v^2 = \frac{k}{\lambda} + c$$

where k and c are constants and v is measured in mm s^{-1} .

- a) Take readings from the graph and draw a suitable table of values.
- b) Using the graph paper **provided on the insert**, plot a straight line graph to test the relationship.
- c) Use your graph to determine the value of k . **[10 marks]**

Total 15 marks

6. (a) A scientist is investigating the rate of conduction, P , of thermal energy through various thicknesses, x , of the material to be used for the walls of a supermarket cold room. The graph in Figure 6 shows the results for various samples of the material, each with an area 0.25 m^2 , for fixed inside and outside temperatures of -5°C and 30°C respectively.

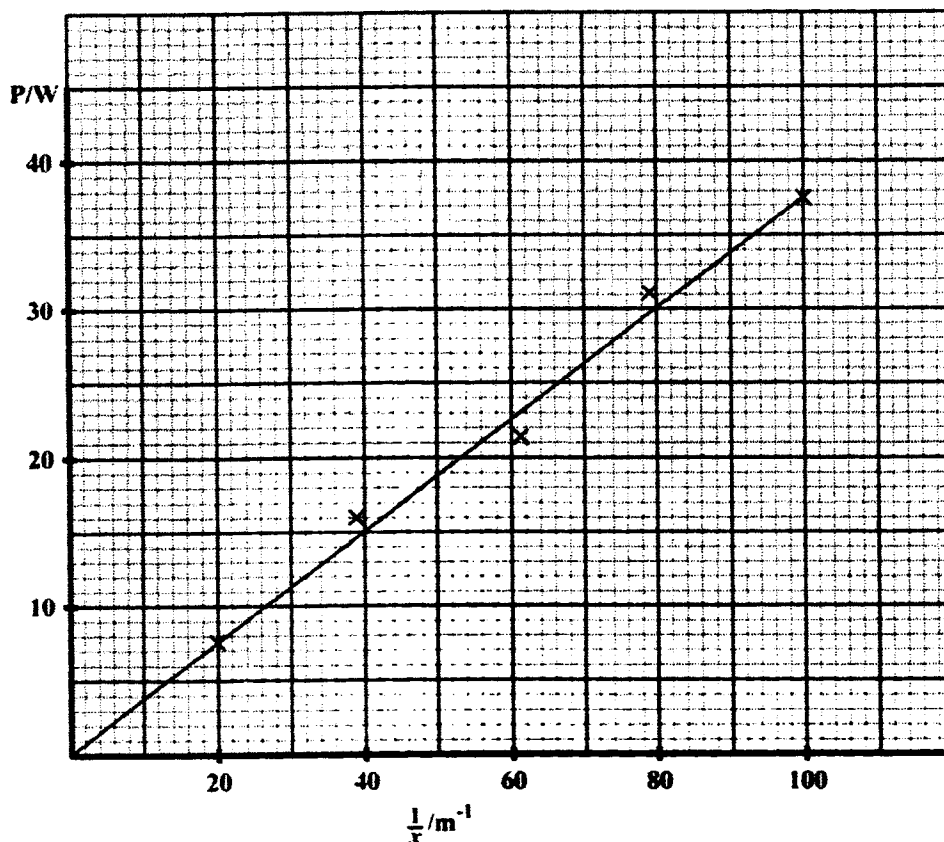


Figure 6

- (i) Write a formula relating P to the inside and outside temperatures of the material, θ_1 and θ_2 , and the thickness, x . [1 mark]
 - (ii) Use the graph to find the rate of flow of thermal energy when $x = 2.5 \text{ cm}$ and when $x = 4.5 \text{ cm}$. [5 marks]
 - (iii) Use the gradient of the graph to find the thermal conductivity of the material. [3 marks]
 - (iv) Sketch the graph which would have been obtained if P were plotted against x instead of $\frac{1}{x}$. [2 marks]
- (b) A packer working in a supermarket goes into the cold room where the temperature is -5°C . If the man's surface area is about 2 m^2 , estimate the net rate of heat loss from his body after just entering the room from outside where the temperature is 30°C . [4 marks]

Total 15 marks

END OF TEST